Recover rules.
Build evidence.
Reuse the method.

AI-assisted discovery → deterministic, auditable Python

01 OBSERVE

Readable baseline + exact errors

02 EXPLAIN

Tools + competing hypotheses

03 VERIFY

Experiments + regression checks

Team NAMA · USI Lugano

Mohammad Javad Amin · Nava Biglar · Malihe Ghasemian · Emad Aghajani

Turn each mismatch into an experiment.

1 DECODE

Dictionary + history

→

2 BASELINE

Readable formula

→

3 EXPLAIN

Conditions + residuals

→

4 TEST

Rank alternatives

→

5 ENCODE

Exact Python

Repeat until the evidence supports the rule

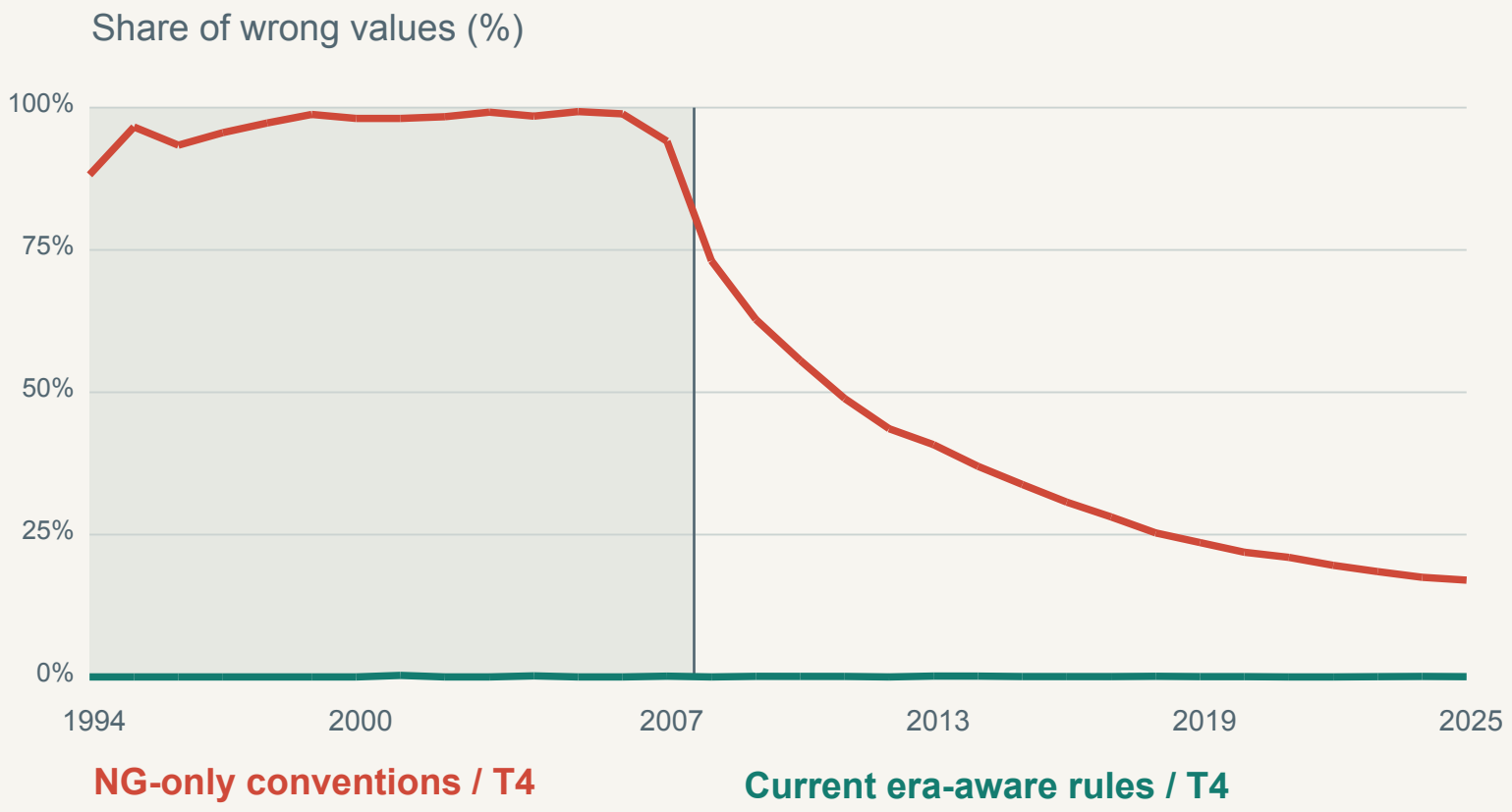
AI + TEAM

Propose, explore, build tools

VERIFICATION

Exact rows + intermediate checks

Find where the system changed.



Local T4 training rows · NG-only: recorded era experiment · current: verified final snapshot

CLUE

Errors cluster by historical era.

EXPERIMENT

Test dated rounding, age and charge conventions.

EXAMPLE 2 / CLEARLY LABELLED ILLUSTRATIVE ARITHMETIC

Use the error shape to propose a formula.

ILLUSTRATION · value 10,000 · annual rate 1% · period 31 days

CANDIDATE A · FIXED MONTH

$10{,}000 \times 1\% \div 12$
8.33

CANDIDATE B · ACTUAL/360

$10{,}000 \times 1\% \times 31 \div 360$
8.61

Unrounded ratio: 31 / 30 = 1.0333…

Residual clue → candidate convention → exact comparison

The search reuses the same engine; each task supplies its candidate choices and scoring function.

Keep investigating when amounts are missing.

Observed witness · BL10002408 / event 8 · Swiss tax residence

GROSS / CHF

5,042.28
Above 5,000

NET / CHF

4,941.43
Below 5,000

LEGACY EVIDENCE

No VST booking

Supports a penalty/threshold hypothesis.

Booking presence constrains applicability. It cannot validate the amount or remittance date.

Sources: Version-2 investigation + current actual-data trace. T7 target amounts are unavailable locally.

New target. Same investigative tools.

REUSE

Residual analysis
Trees + change points
Candidate search + exact checks

ADAPT

Schema + relationships
Candidate features and formulas
Reference evidence

99.735–100%

Exact rows · T0–T4 · local training targets

Human-guided discovery.
Evidence-backed implementation.

T7 evaluated baseline: payout 1/20; annual 15/20. Current TRACE candidate: organizer result pending.

The reusable toolkit has explicit inputs.

Rule finder	features + mismatch flags	Readable conditions to investigate
Change points	dates + mismatch flags	Ranked historical boundaries
Residuals	calculated + reference values	Rounding and ratio signatures
Convention search	candidate options + score(choice)	Ranked candidate combinations
Weak labels	predicted applicability + bookings	Agreement / disagreement cases

Stack: Python · pandas · scikit-learn · matplotlib · Decimal

Keep local evidence separate from hidden scores.

LOCAL TRAINING TARGETS

T0 Commission	225,848 / 225,858	**99.9956%**
T1 Due dates	15,028 / 15,028	**100%**
T2 Maturity	3,011 / 3,019	**99.7350%**
T3 Surrender	6,137 / 6,153	**99.7400%**
T4 Statement	34,124 / 34,145	**99.9385%**

MAINTAINABILITY

82.5
Local scorer

REGRESSION

27 T7 tests
29 T6 checks

T5: 55 flags; hidden F1 unavailable. T7 evaluated baseline: 1/20 payout, 15/20 annual.

Latest TRACE overlay: seven stamp-only payout changes; all 6,381 annual rows unchanged vs updated main.

Show what supported each decision.

OBSERVED

Training diagnostics, bookings, version history and organizer counts.

ILLUSTRATIVE

The charge arithmetic example explains a signature; it is not a policy trace.

AI DISCLOSURE

Claude Code + Codex assisted investigation, coding and review. The team directs and evaluates.

BOUNDARY

Reusable tools need new adapters and hypotheses. Hidden T7 accuracy remains unresolved.